24	**D**	Since the magnetic force on the ions provides the centripetal force to enable the ion to in circular path in the magnetic field $Bqv = \dfrac{mv^2}{r}$ $r = \dfrac{v}{B}\dfrac{m}{q}$ Since the magnetic flux density *B* and speed of the ion *v* remain unchanged. $r \propto \dfrac{m}{q}$, $r_1 < r_2$, $\dfrac{m_1}{q_1} < \dfrac{m_2}{q_2}$ $\dfrac{q_2}{m_2} < \dfrac{q_1}{m_1}$
25	**D**	